> $12+4-5$

$$11 \tag{1}$$

> 2^{10}

$$1024 \tag{2}$$

> $\sin(0.1)$

$$0.09983341665 \tag{3}$$

> $(a+b)(a-b)$

$$a(a-b)+b(a-b) \tag{4}$$

> $f := x \rightarrow 3x^3+2x^2-5:$

> $diff(f(x), x)$

$$9x^2+4x \tag{5}$$

> $f := x \rightarrow \mathrm{sqrt}(1+x^4):$

> $diff(f(x), x)$

$$\frac{2x^3}{\sqrt{x^4+1}} \tag{6}$$

> $f := x \rightarrow \exp(x) \cdot \sin(x) \cdot \cos(x):$

> $diff(f(x), x)$

$$\mathrm{e}^x \sin(x)\cos(x)+\mathrm{e}^x \cos(x)^2-\mathrm{e}^x \sin(x)^2 \tag{7}$$

> $simplify((a+b)(a-b))$

$$a(a-b)+b(a-b) \tag{8}$$

> $expand((a+b)\cdot(a-b))$

$$a^2-b^2 \tag{9}$$

> $simplify((a-b)\cdot(a+b))$

$$a^2-b^2 \tag{10}$$

> $int(3x^3+2x^2-5, x=0..1)$

$$-\frac{43}{12} \tag{11}$$

> $int\left(\frac{1}{x^2}, x=0..\mathrm{infinity}\right)$

$$\infty \tag{12}$$

> $int(\exp(-x^2), x=-\mathrm{infinity}..\mathrm{infinity});$

$$\sqrt{\pi} \tag{13}$$

> $limit\left(\frac{\sin(x)}{x}, x=0\right)$

$$1 \tag{14}$$

> $limit\left(\frac{(x^3+3x^2-5)}{2x^3-7x}, x=\mathrm{infinity}\right)$

$$\frac{1}{2} \tag{15}$$

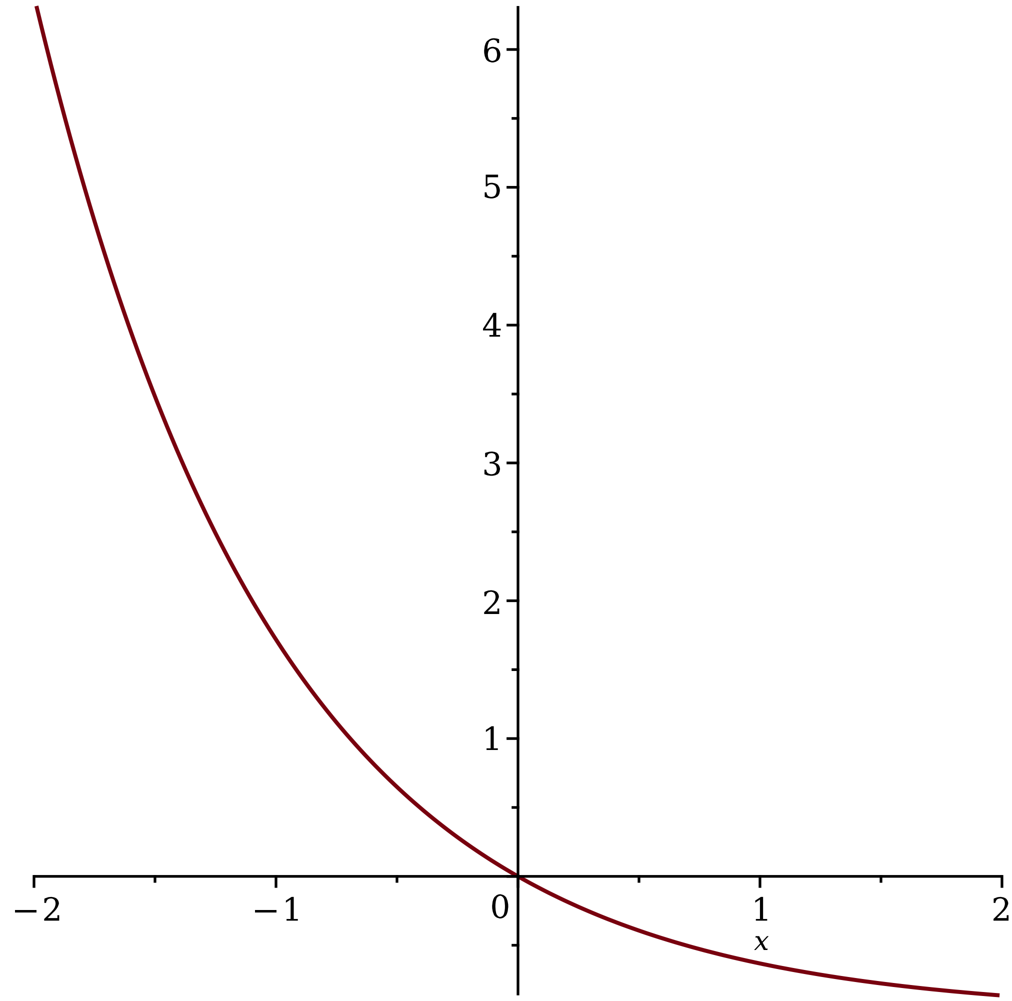
> $limit\left(\frac{(\cos(x)+1)}{x-\mathrm{Pi}}, x=\mathrm{Pi}\right)$

$$0 \tag{16}$$

> $f := x \rightarrow \exp(-x) - 1:$

> $with(plots):$

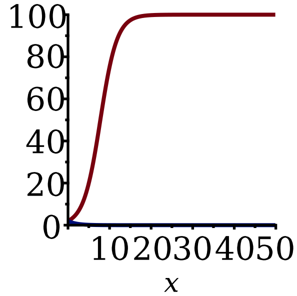
> $plot(f(x), x=-2..2)$



> $f := (x, r) \rightarrow \frac{(200 \cdot \exp(r \cdot x))}{2(\exp(r \cdot x) - 1) + 100}:$

> $with(plots):$

> $plot([f(x, 0.5), f(x, -0.5)], x=0..50), color=[red, blue]$

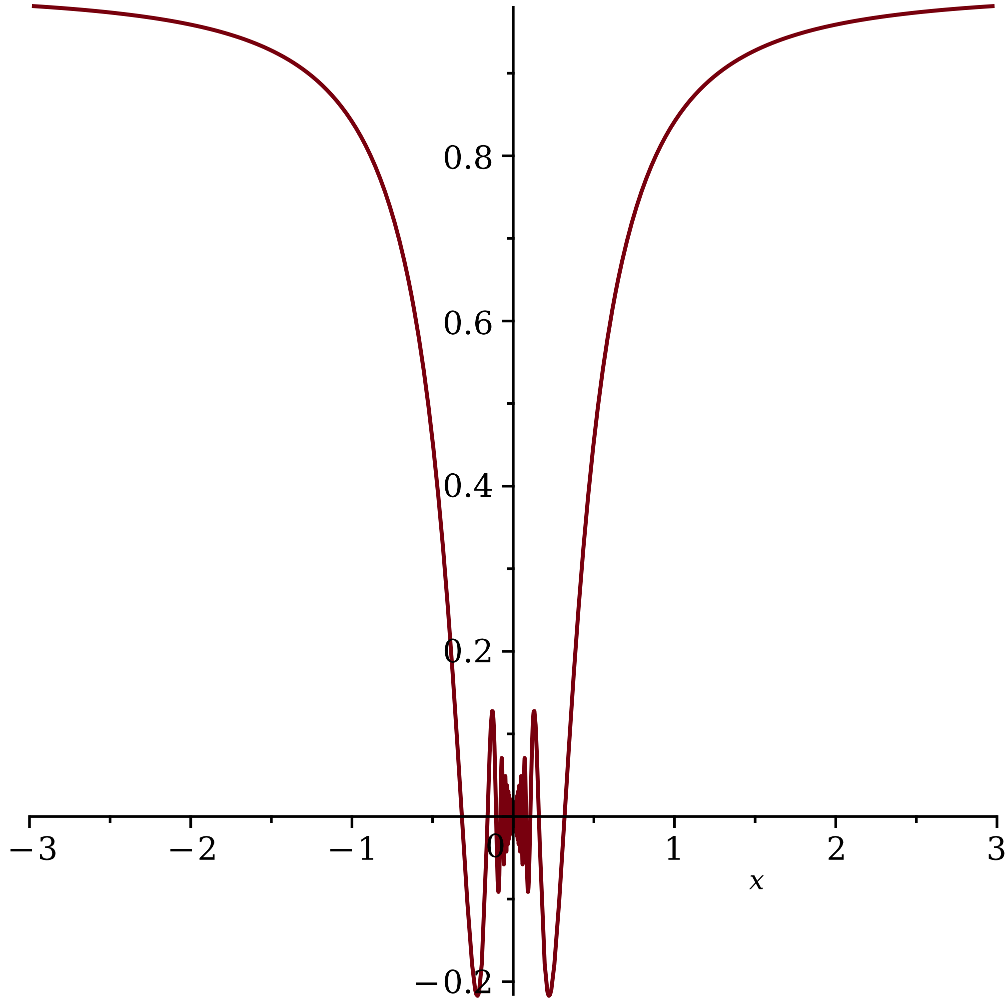


$, color=[red, blue]$

> $f := x \to x \cdot \sin\left(\frac{1}{x}\right)$:

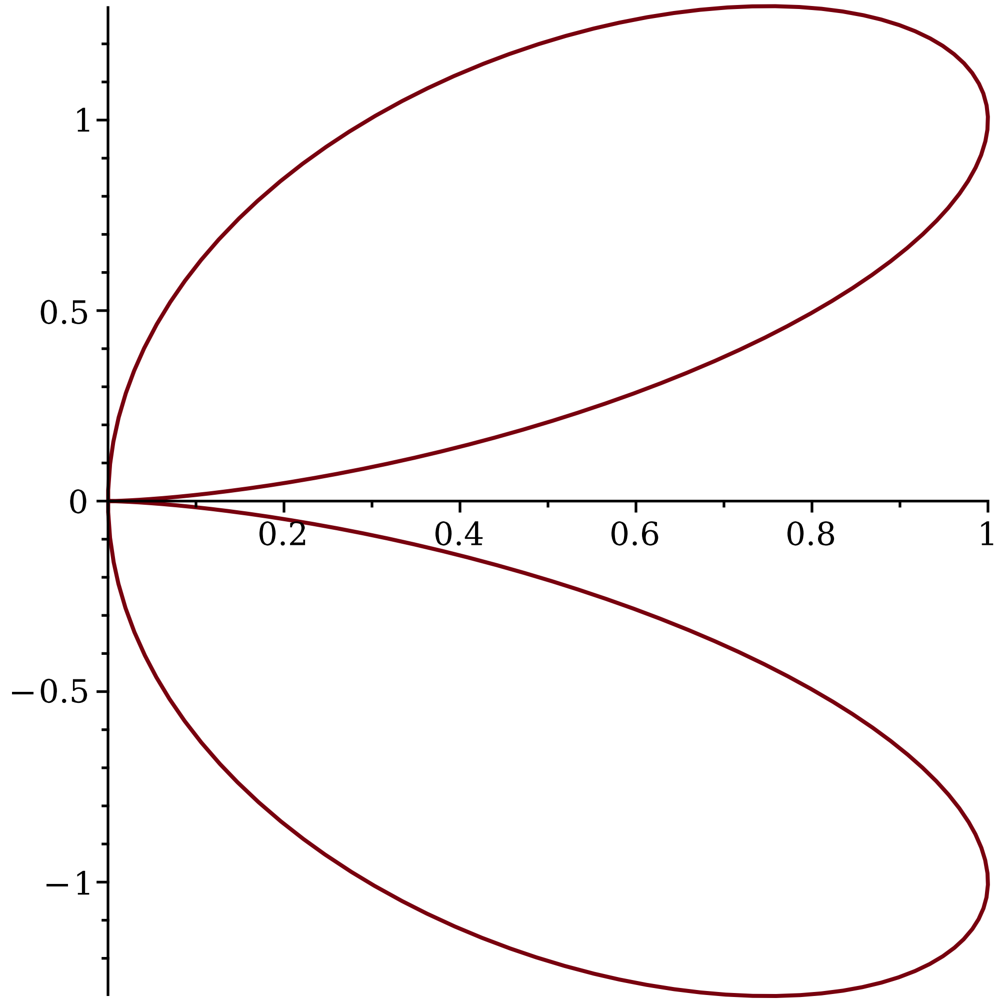
> $with(plots)$:

> $plot(f(x), x = -3..3)$



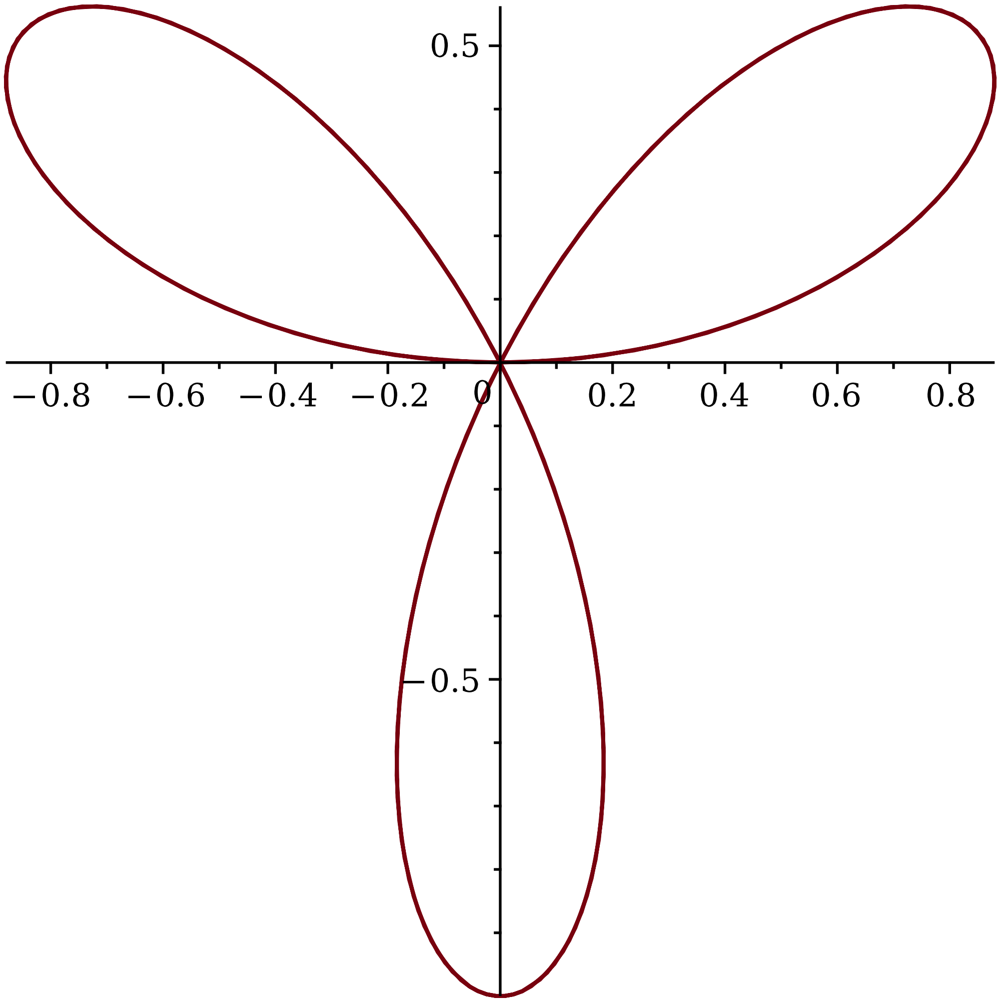
> $with(plots)$:

> $plot([(1 - \cos(t) \cdot \cos(t)), (1 - \cos(t)) \cdot \sin(t), t = 0..2\,\mathrm{Pi}])$



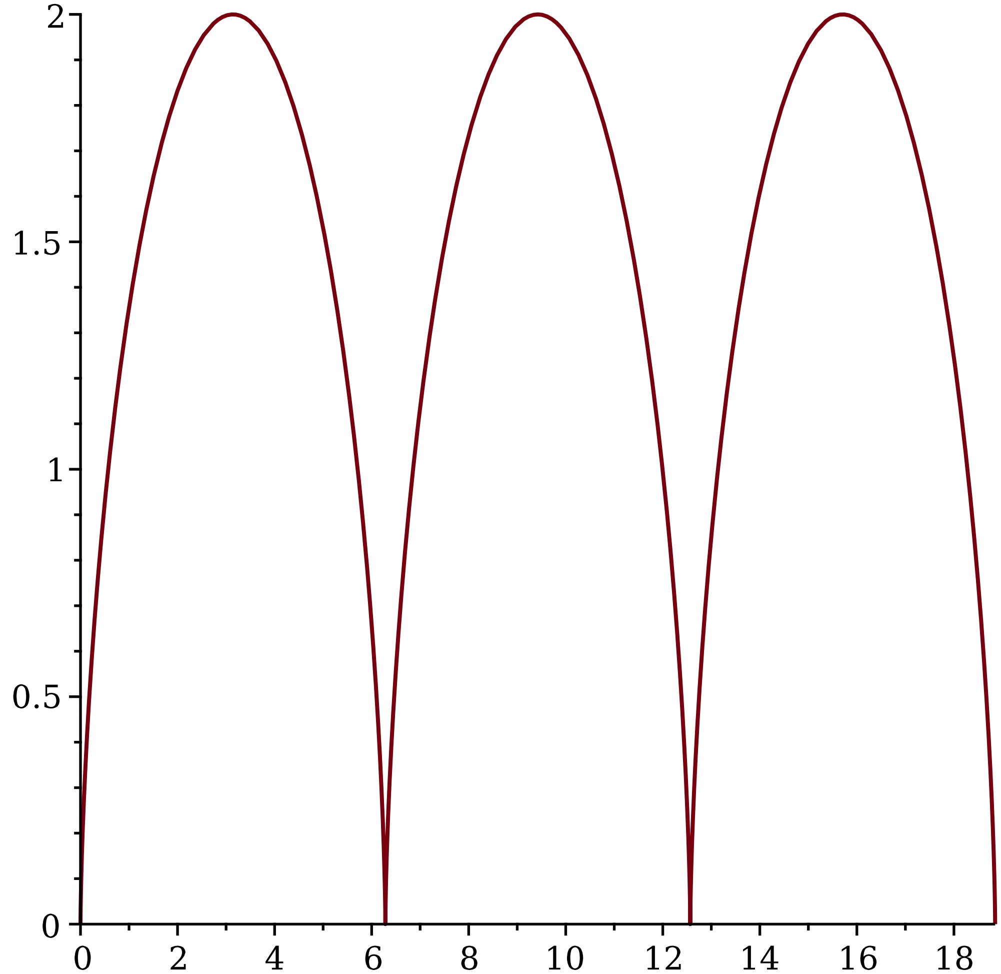
> $with(plots):$

> $plot([\sin(3\cdot t)\cos(t), \sin(3\cdot t)\sin(t), t = 0..2\cdot \mathrm{Pi}])$



> $with(plots)$:

> $plot([t - \sin(t), 1 - \cos(t), t = 0..6 \cdot \mathrm{Pi}])$



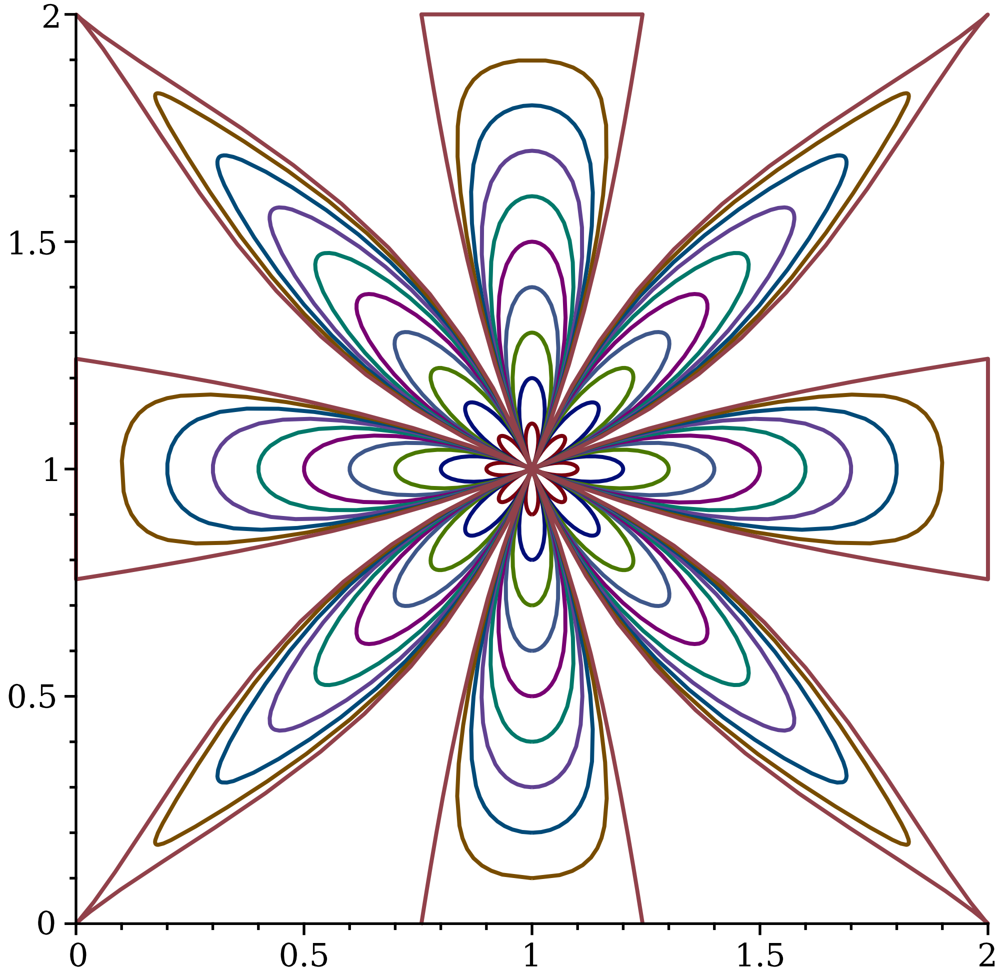
> $f := (t, s) \rightarrow 1 - \left(\frac{s \cdot \cos(4\,t) \cdot \cos(t)}{\operatorname{sqrt}(1 - s^2 \cdot \cos^2(4\,t) \cdot \sin^2(t))} \right) :$

> $x := (t, s) \rightarrow f\left(t - \frac{\mathrm{Pi}}{2}, s\right) :$

> $y := (t, s) \rightarrow f(t, s) :$

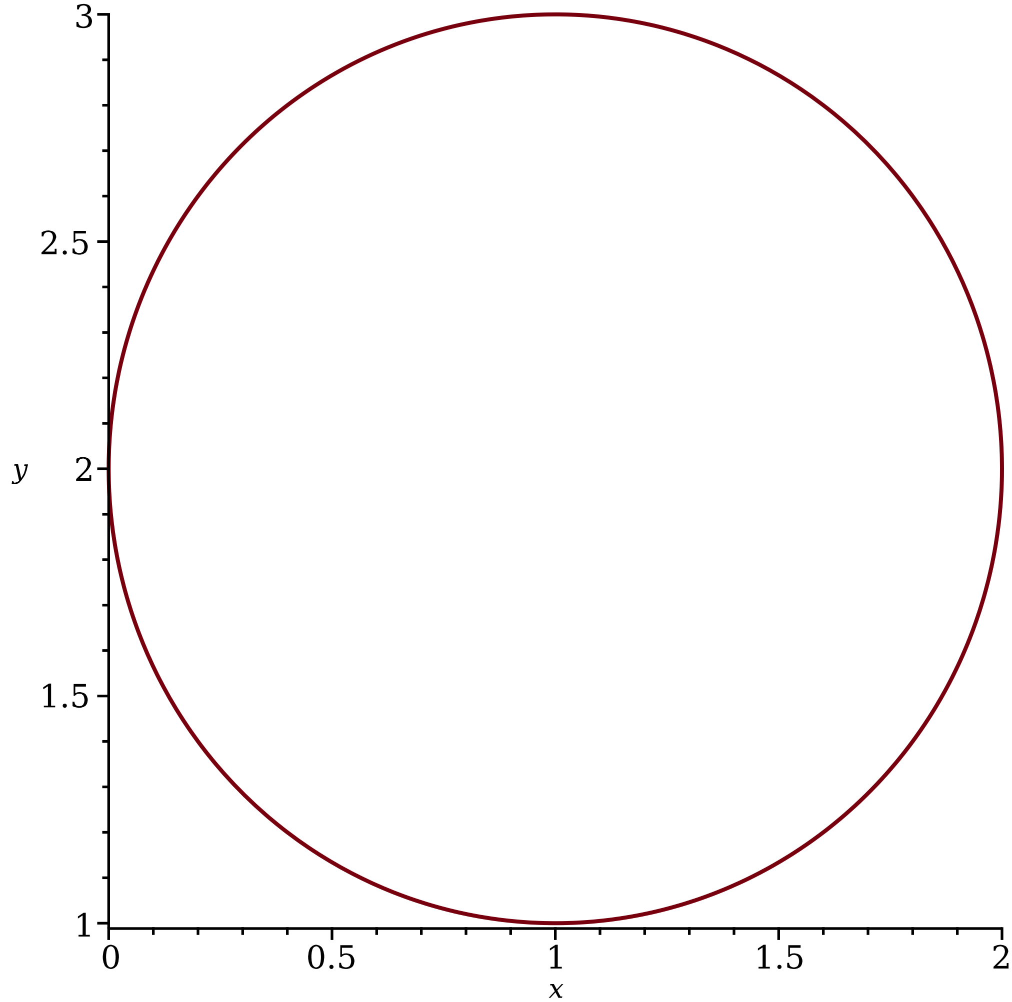
> $list\_xy := \left[x\left(t, \frac{s}{10}\right), y\left(t, \frac{s}{10}\right), t = 0\,..2 \cdot \mathrm{Pi}\right] \$ s = 1\,..10 :$

> $plot([list\_xy])$



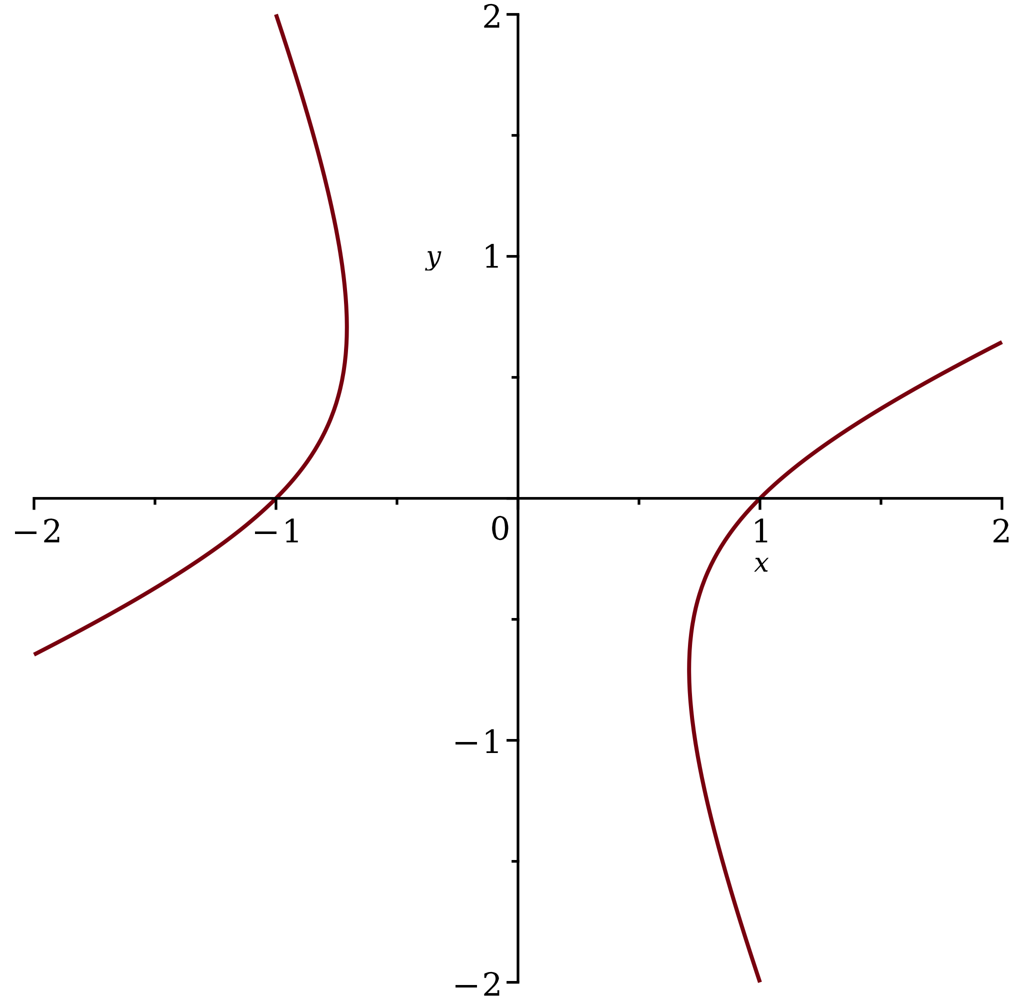
> $with(plots):$

> $implicitplot(x^2 + y^2 - 2\,x - 4\,y + 4 = 0)$



> $with(plots):$

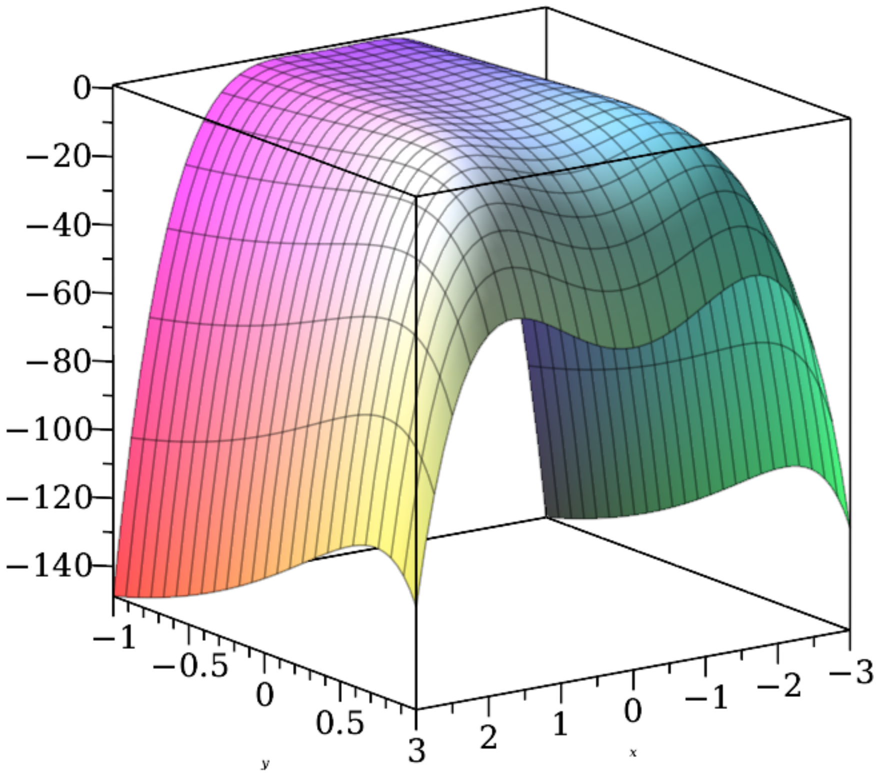
> $implicitplot((x-y)^2-2y^2=1)$



> $z := (x, y) \rightarrow 4\,x^2 \cdot \exp(y) - 2\,x^4 - \exp(4\,y)$:

> $with(plots)$:

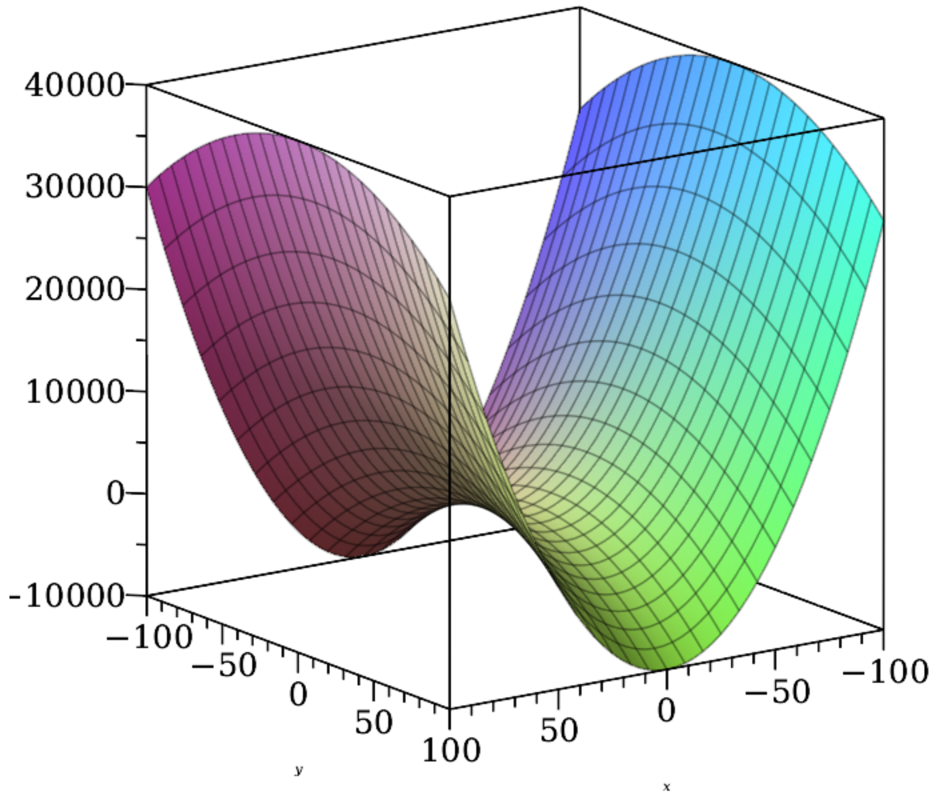
> $plot3d(z(x, y), x = -3..3, y = -1..1, axes = boxed)$



> $z := (x, y) \to 4\,x^2 - y^2$:

> $with(plots)$:

> $plot3d(z(x, y), x = -100..100, y = -100..100, axes = boxed)$



> $with(linalg):$

> $A := matrix([[1, 2, -1], [0, 1, 0], [3, -1, 2]]):$

> $B := matrix([[1, 2, 3], [1, 1, 2], [2, 1, 1]]):$

> $C := matrix([[2, 1, 1], [0, 1, -1], [4, 2, 2]]):$

> $evalm(2 \cdot A - B\&{*}C);$

$$\begin{bmatrix} -12 & -5 & -7 \\ -10 & -4 & -4 \\ -2 & -7 & 1 \end{bmatrix}$$

> $evalm(A^{(-1)});$

$$\begin{bmatrix} \frac{2}{5} & -\frac{3}{5} & \frac{1}{5} \\ 0 & 1 & 0 \\ -\frac{3}{5} & \frac{7}{5} & \frac{1}{5} \end{bmatrix} \tag{18}$$

> *eigenvects*(C);

$$\left[0, 1, \left\{\left[\begin{array}{ccc} -1 & 1 & 1 \end{array}\right]\right\}\right], \left[2, 1, \left\{\left[\begin{array}{ccc} 1 & -2 & 2 \end{array}\right]\right\}\right], \left[3, 1, \left\{\left[\begin{array}{ccc} -1 & 1 & -2 \end{array}\right]\right\}\right] \tag{19}$$

> *eigenvals*(C)

$$0, 3, 2 \tag{20}$$

>